

```
import pandas as pd
df = pd.read_csv('marketing_and_sales_data_evaluate.csv.csv')
```

```
df.head()
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

```
#checking the dataset structure#
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0    TV              4562 non-null   float64
1    Radio           4568 non-null   float64
2    Social_Media    4566 non-null   float64
3    Sales           4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

```
#checking for missing values#
df.isnull().sum()
```

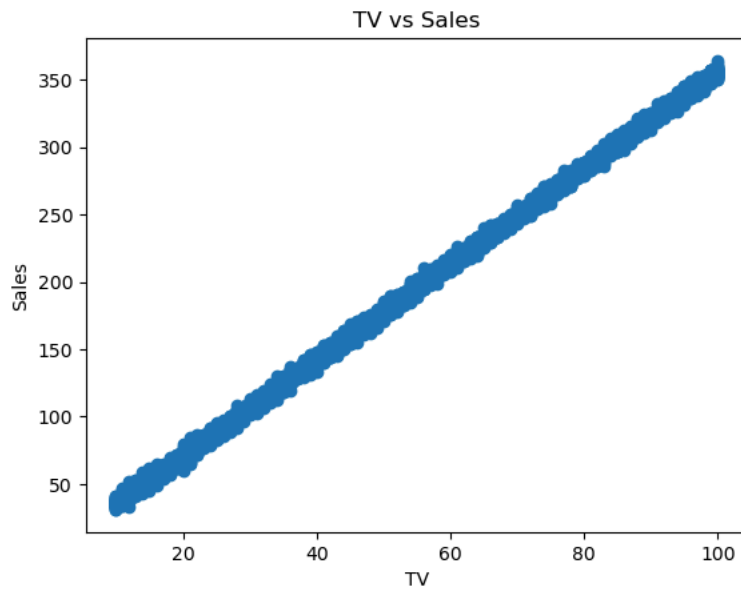
```
TV          10
Radio        4
Social_Media 6
Sales        6
dtype: int64
```

```
#removing missing values#
df = df.dropna()
```

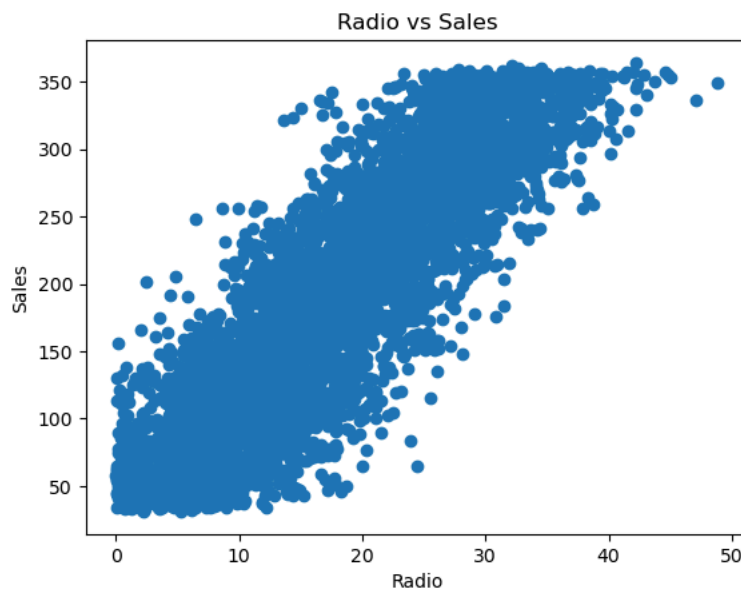
```
# performing Exploratory Data Analysis(EDA)#
df.describe()
```

	TV	Radio	Social_Media	Sales
count	4546.000000	4546.000000	4546.000000	4546.000000
mean	54.062912	18.157533	3.323473	192.413332
std	26.104942	9.663260	2.211254	93.019873
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.555355	1.530822	112.434612
50%	53.000000	17.859513	3.055565	188.963678
75%	77.000000	25.640603	4.804919	272.324236
max	100.000000	48.871161	13.981662	364.079751

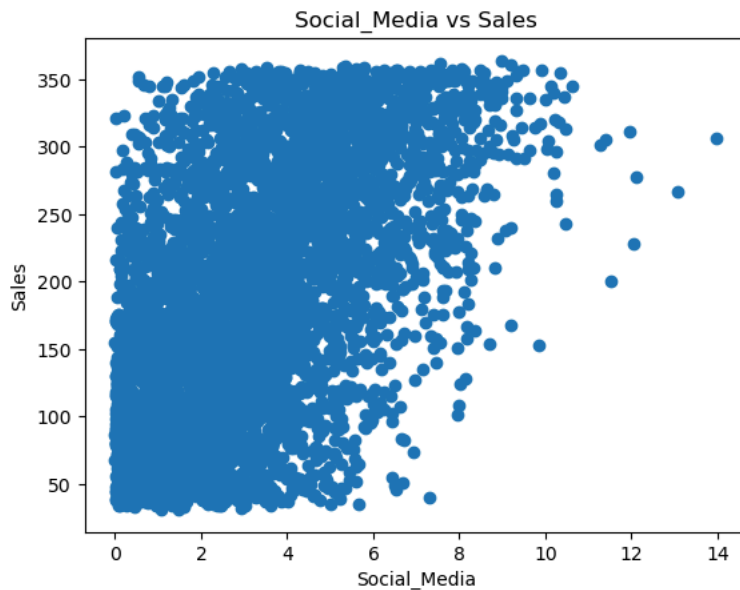
```
# visualizing relationship#
import matplotlib.pyplot as plt
plt.scatter(df['TV'], df['Sales'])
plt.xlabel('TV')
plt.ylabel('Sales')
plt.title('TV vs Sales')
plt.show()
```



```
plt.scatter(df['Radio'], df['Sales'])  
plt.xlabel('Radio')  
plt.ylabel('Sales')  
plt.title('Radio vs Sales')  
plt.show()
```



```
plt.scatter(df['Social_Media'], df['Sales'])  
plt.xlabel('Social_Media')  
plt.ylabel('Sales')  
plt.title('Social_Media vs Sales')  
plt.show()
```



```
# Identifying the most correlated variable with sales#
correlation = df.corr()
print(correlation['Sales'])
```

```
TV          0.999497
Radio       0.868638
Social_Media 0.527446
Sales       1.000000
Name: Sales, dtype: float64
```

```
#Building the OLS Regression Model#
import statsmodels.api as sm
X = df['TV']
Y = df['Sales']
X = sm.add_constant(X)
model = sm.OLS(Y,X).fit()
print(model.summary())
```

```
=====
                        OLS Regression Results
=====
```

Dep. Variable:	Sales	R-squared:	0.999
Model:	OLS	Adj. R-squared:	0.999
Method:	Least Squares	F-statistic:	4.517e+06
Date:	Sat, 20 Jun 2026	Prob (F-statistic):	0.00
Time:	10:59:09	Log-Likelihood:	-11366.
No. Observations:	4546	AIC:	2.274e+04
Df Residuals:	4544	BIC:	2.275e+04
Df Model:	1		
Covariance Type:	nonrobust		

```
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	-0.1325	0.101	-1.317	0.188	-0.330	0.065
TV	3.5615	0.002	2125.272	0.000	3.558	3.565

```
=====
```

Omnibus:	0.052	Durbin-Watson:	1.998
Prob(Omnibus):	0.974	Jarque-Bera (JB):	0.031
Skew:	-0.001	Prob(JB):	0.985
Kurtosis:	3.012	Cond. No.	138.

```
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```

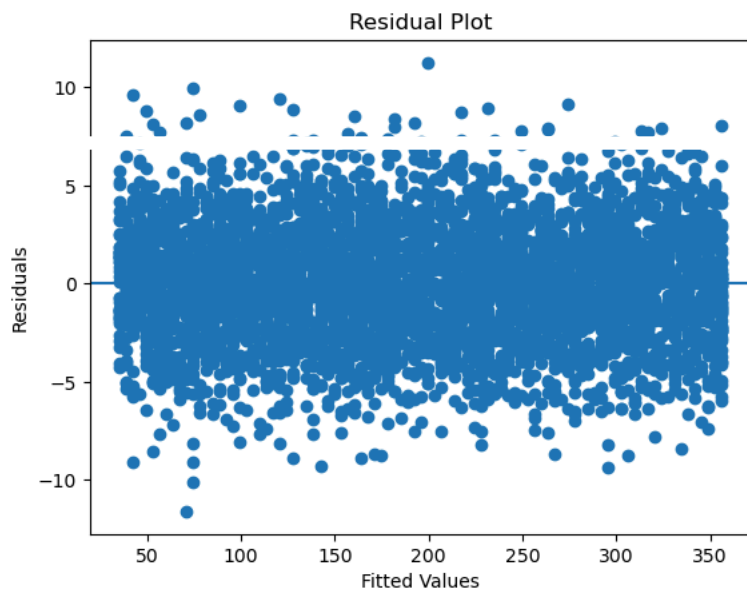
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

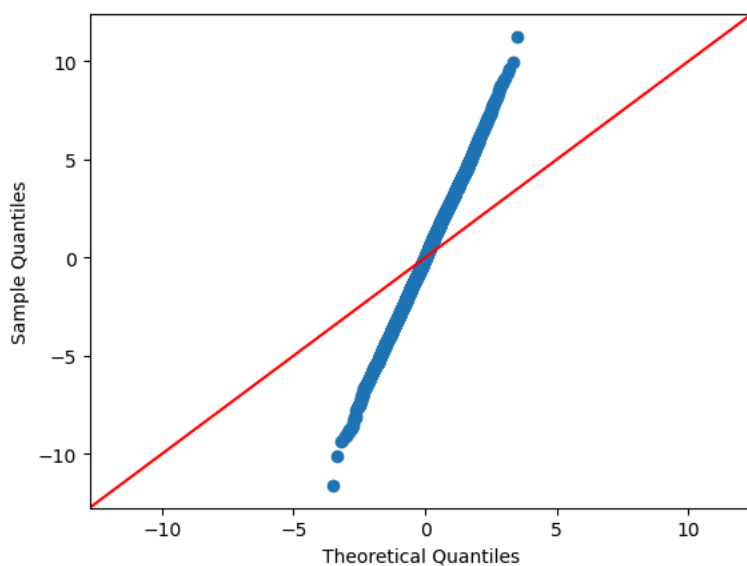
Regression Equation Sales = -0.126+3.561(TV) R-Squared R2 =0.998995 P-values Variable P-value TV <0.001 Constant 0.209

```
#Showing diagnostic plots#
plt.scatter(model.fittedvalues, model.resid)
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residual6 Plot')
```

```
nlt.axhline(v=0)
```



```
#Q-Q Plot#  
sm.qqplot(model.resid, line='45')  
plt.show()
```



Business recommendation#

Based on the analysis, TV advertising demonstrates the strongest relationship with sales and explains approximately 99.9% of sales variations. Therefore the company should allocate a larger proportion of its marketing budget to TV campaigns to maximize return on investment (ROI), while using Radio and Social Media as supporting channels.

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